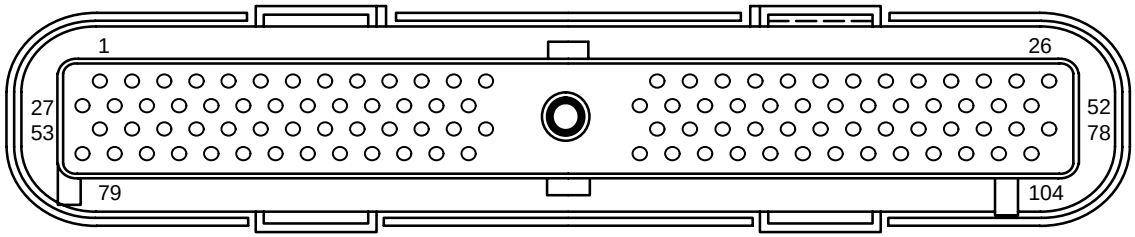
C175 (GY)

12A581

Powertrain Control Module (PCM) (12A650)



F104001

* 4.2L
** 4.6L
*** 4.2L, 4.6L
† 5.4L Natural Gas Vehicle (NGV)
†† 6.8L
††† 7.3L
‡ except 4.6L, 6.8L
‡‡ except 4.2L
‡‡‡ except 5.4L Natural Gas Vehicle (NGV)
◆ except diesel

Pin	Circuit	Circuit function
1	◆1025 (OG/YE)	Coil On Plug (COP) 6 (12029), control
	*315 (VT/OG)	Shift solenoid B
2	658 (PK/LG)	Malfunction Indicator Lamp (MIL), control
3	570 (BK/WH)	Ground
	*1144 (YE/BK)	Digital Transmission Range (DTR) sensor, TR1
	†††3996 (RD/YE)	Manifold intake air heater relay, switched output
4	‡828 (VT/LB)	Voltage supplied at all times (overload protected)
	†††284 (OG)	not used
5	†††162 (LG/RD)	Parking brake switch (15A851)
6	237 (OG/YE)	Shift solenoid A
7	–	not used
8	*1022 (DB/YE)	Intake Manifold Runner Control (IMRC) module, monitor
	†††1278 (WH/VT)	output, to, Glow plug relay
9	‡29 (YE/WH)	Fuel pump/fuel gauge sender, Low fuel indicator
	†††1279 (WH)	Fuel pump/fuel gauge sender
10	†††308 (RD/OG)	Idle validation switch, control
11	315 (VT/OG)	Shift solenoid B
12	911 (WH/LG)	Transmission Control Indicator Lamp (TCIL)
	*29 (YE/WH)	Fuel pump/fuel gauge sender

Pin	Circuit	Circuit function
13	107 (VT)	Module programming signal
14	781 (OG/LB)	to 4L switch
15	915 (PK/LB)	SCP Bus –
16	914 (TN/OG)	SCP Bus +
17	†††1012 (OG/BK)	Digital Transmission Range (DTR) sensor (7F293), TR1
18	–	not used
19	†††648 (WH/PK)	signal
20	924 (BN/OG)	Powertrain Control Module (PCM) (12A650), Coasting clutch solenoid
21	349 (DB)	Crankshaft position sensor (6C315) +
	†††795 (DG)	Camshaft position sensor (6B288), to, Powertrain Control Module (PCM) (12A650)
22	350 (GY)	Crankshaft position sensor (6C315) –
23	††570 (BK/WH)	Ground
24	570 (BK/WH)	Ground
	†††837 (YE/BK)	Accelerator pedal position sensor, to, Powertrain Control Module (PCM) (12A650)
25	875 (BK/LB)	Ground
26	◆1024 (LG/WH)	Coil On Plug (COP) 1 (12029), control
	*526 (DB/LG)	PCM, to, Ignition coil (12029)

Pin	Circuit	Circuit function
27	◆1021 (LG/YE)	Coil On Plug (COP) 5 (12029), control
	††1142 (WH/RD)	Coil On Plug (COP) 10, control
	*237 (OG/YE)	Shift solenoid A
28	†††643 (RD)	indicator, Water–in–fuel sensor
29	224 (TN/WH)	PCM, to transmission overdrive cancel switch
30	†††553 (VT/LB)	Exhaust Back Pressure (EBP) sensor, to, Powertrain Control Module (PCM) (12A650)
31	†††810 (RD/LG)	Deactivator switch, input
32	311 (DG/VT)	Knock sensor (12A699) –
33	†††676 (PK/OG)	Vehicle Speed Sensor (VSS) –
34	‡‡1144 (YE/BK)	Digital Transmission Range (DTR) sensor, TR1
	†††1277 (WH/LG)	not used
35	‡‡‡392 (RD/LG)	Heated Oxygen Sensor (HO2S) #12 (9G444), input
	†††1183 (WH/YE)	Generator, Power
36	968 (TN/LB)	Mass Air Flow (MAF) sensor (12B579), signal, return
	†††1280 (GY/RD)	Fuel heater, to, Powertrain Control Module (PCM) (12A650)
37	923 (OG/BK)	Transmission Fluid Temperature (TFT) sensor, input
38	†††354 (LG/RD)	Engine Oil Temperature (EOT) sensor, input
39	743 (GY)	Intake Air Temperature (IAT) sensor (12A697), input
40	238 (DG/YE)	Fuel pump relay, control
	†††787 (PK/BK)	Fuel pump, monitor
41	347 (BK/YE)	A/C pressure switch, signal
	**439 (TN/LG)	A/C pressure switch, signal

Pin	Circuit	Circuit function
42	*1023 (DB/LG)	Intake Manifold Runner Control (IMRC) module, to, Powertrain Control Module (PCM) (12A650)
	†††318 (GY/RD)	Exhaust Pressure Regulator (EPR), to, Powertrain Control Module (PCM) (12A650)
	††1101 (GY/RD)	Fuel injector 10 (9F593), control
43	–	not used
44	–	not used
45	1105 (YE/LB)	Engine Coolant Temperature (ECT) sender, to, Temperature, indicator
	†††683 (VT/LB)	Cruise set, indicator, control
46	**367 (BN)	Intake Manifold Runner Control (IMRC) module
	††679 (GY/BK)	Vehicle Speed Sensor (VSS) +
47	360 (BN/PK)	EGR Vacuum Regulator (EVR), control
48	648 (WH/PK)	Tachometer, signal
	†††818 (GY/WH)	Injector Driver Module (IDM), to, Powertrain Control Module (PCM) (12A650)
49	1145 (LB/BK)	Digital Transmission Range (DTR) sensor, TR2
	†††146 (WH/PK)	Digital Transmission Range (DTR) sensor, TR2
50	1146 (LG/RD)	Digital Transmission Range (DTR) sensor, TR4
	†††145 (GY/BK)	Digital Transmission Range (DTR) sensor, TR4
51	570 (BK/WH)	Ground
52	◆1028 (WH/PK)	Coil On Plug (COP) 3 (12029), control
	††1021 (LG/YE)	Coil On Plug (COP) 5 (12029), control
	*527 (RD/LB)	PCM, to, Ignition coil (12029)
53	◆1030 (DG/VT)	Coil On Plug (COP) 4 (12029), control
	††1027 (PK/LB)	Coil On Plug (COP) 7 (12029), control
54	480 (VT/YE)	Torque Converter Clutch (TCC) solenoid
55	37 (YE)	Voltage supplied at all times (overload protected)

Pin	Circuit	Circuit function
56	191 (LG/BK)	Powertrain Control Module (PCM) (12A650), EVAP canister purge valve
57	310 (YE/RD)	Knock sensor (12A699) –
58	†††679 (GY/BK)	Vehicle Speed Sensor (VSS) +
59	970 (DG/WH)	Turbine Shaft Speed (TSS) sensor
60	74 (GY/LB)	Heated Oxygen Sensor (HO2S) #11 (9F472), input
	†††1185 (YE)	Generator, Power
61	***393 (VT/LG)	Heated Oxygen Sensor (HO2S) #22 (9G444), input
	†††151 (LB/BK)	Speed control switch, input
62	◆791 (RD/PK)	Fuel tank pressure transducer sensor (9C968)
	†1164 (LB)	Engine controls, Gas temperature sensor (CNG), Temperature
63	†141 (RD/PK)	Injector pressure sensor, except 1,9 TDI unit injector engine
64	199 (LB/YE)	Digital Transmission Range (DTR) sensor, TR3A
65	‡‡‡352 (BN/LG)	Differential Pressure Feedback EGR (DPFE) sensor, input
	†††796 (LB)	Camshaft position sensor (6B288), signal, return
66	1102 (YE/LG)	Cylinder–head temperature sensor (6G004), to, Powertrain Control Module (PCM) (12A650)
	†††439 (TN/LG)	Modified vehicle power, to, Powertrain Control Module (PCM) (12A650)
67	91 (VT/WH)	Canister vent
	905 (GY/LB)	Canister vent
68	◆679 (GY/BK)	Vehicle Speed Sensor (VSS) +
	††1100 (GY/BK)	Fuel injector 9 (9F593), control
69	**883 (BK/LB)	Voltage supplied at all times (overload protected)
	*331 (PK/YE)	A/C clutch relay, control
70	†††464 (BK/PK)	Low coolant indicator, control

Pin	Circuit	Circuit function
71	361 (RD)	Voltage supplied in Start and Run (overload protected)
72	561 (TN/RD)	Fuel injector 7, control
73	559 (TN/BK)	Fuel injector 5, control
74	557 (BN/YE)	Fuel injector 3, control
75	555 (TN)	Fuel injector 1, control
76	570 (BK/WH)	Ground
77	570 (BK/WH)	Ground
78	◆1027 (PK/LB)	Coil On Plug (COP) 7 (12029), control
	*528 (PK/WH)	PCM, to, Ignition coil (12029)
	††1026 (PK/WH)	Coil On Plug (COP) 2 (12029), control
79	1029 (WH/RD)	Coil On Plug (COP) 8 (12029), control
	*911 (WH/LG)	Transmission Control Indicator Lamp (TCIL)
	†††358 (LG/BK)	Manifold Absolute Pressure (MAP) sensor, to, Powertrain Control Module (PCM) (12A650)
80	926 (LB/OG)	Fuel pump relay, control
	†††814 (WH/BK)	Injector Driver Module (IDM) power relay, control
81	925 (WH/YE)	Electronic Pressure Control (EPC) solenoid, control
82	††1141 (YE/BK)	Coil On Plug (COP) 4 (12029), control
83	264 (WH/LB)	Idle Air Control (IAC) valve (9F715), control
	†††552 (YE/RD)	Injection Pressure Regulator (IPR), to, Powertrain Control Module (PCM) (12A650)
84	136 (DB/YE)	Powertrain Control Module (PCM) (12A650) – Output Shaft Speed (OSS) sensor (7H103)
	†††970 (DG/WH)	Turbine Shaft Speed (TSS) sensor
85	795 (DG)	Camshaft position sensor (6B288), to, Powertrain Control Module (PCM) (12A650)
86	–	not used
87	94 (RD/BK)	Heated Oxygen Sensor (HO2S) #21 (9F472), input
	†††812 (DB/LG)	Injection Control Pressure (ICP) sensor

Pin	Circuit	Circuit function
88	967 (LB/RD)	Mass Air Flow (MAF) sensor (12B579), output
89	355 (GY/WH)	Throttle Position Sensor (TPS) (9B989), input
90	351 (BN/WH)	Reference, voltage
91	359 (GY/RD)	signal, return
92	511 (LG)	Brake pedal position switch (13480), to stoplamps
93	387 (RD/WH)	Heated Oxygen Sensor (HO2S) #11 (9F472), heater
94	388 (YE/LB)	Heated Oxygen Sensor (HO2S) #21 (9F472), heater
	†††926 (LB/OG)	Fuel pump relay, control
95	‡‡‡389 (WH/BK)	Heated Oxygen Sensor (HO2S) #12 (9G444), heater
	†††821 (BN/OG)	fuel delivery command, signal, output
96	***390 (TN/YE)	Heated Oxygen Sensor (HO2S) #22 (9G444), heater
	†††817 (YE/LB)	cylinder identification, signal, output
97	361 (RD)	Voltage supplied in Start and Run (overload protected)
98	‡‡562 (LB)	Fuel injector 8, control
	†††462 (VT)	Manifold intake air heater relay, control
99	560 (LG/OG)	Fuel injector 6, control
100	558 (BN/LB)	Fuel injector 4, control
	†††29 (YE/WH)	Fuel pump/fuel gauge sender
101	556 (WH)	Fuel injector 2, control
	†††1086 (VT/OG)	Glow plug relay, to, Powertrain Control Module (PCM) (12A650)
102	††1030 (DG/VT)	Coil On Plug (COP) 4 (12029), control
103	††570 (BK/WH)	Ground
104	◆1026 (PK/WH)	Coil On Plug (COP) 2 (12029), control
	††1028 (WH/PK)	Coil On Plug (COP) 3 (12029), control